Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

Perfomax *AI Advisor* **2026–present**

National University of Singapore *Postdoc under Kostya Novoselov* **2022–present**

Leadership

- Program Chair for the AI4X 2025/6 conferences
- Main organizer of the ICLR 2025 Workshop on Machine Learning Multiscale Processes

Wyckoff Transformer for generating symmetric crystals ICML 2025 · GitHub

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on PyPI
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

CERN [Yandex → HSE University] *Researcher* **2014–2022**

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Generative models uncertainty estimation paper · conference · code 1 · code 2

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation paper 1 · paper 2 · paper 3 · code

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Education

HSE University & Sapienza Università di Roma **2016–2020**
PhD in Computer Science and Physics (Double Degree)
Supervisors: Andrey Ustyuzhanin & Barbara Sciascia
Thesis: Machine Learning for particle identification in the LHCb detector

Product Management course at Yandex **2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's-level CS programme
Algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics **2014–2016**
Optimisation of data processing of the LHCb experiment.
BS in Physics **2010–2014**
Wave-packet molecular dynamics of electrons in nonideal plasma.

International Junior Science Olympiad (IJSO) **Korea, 2008**
Silver medal

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a $3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Lecturer in the award-winning Coursera Advanced Machine Learning specialisation
- Delivered more than 20 conference talks.

Service

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

- **ML & AI** — Structured, non-text, domain-specific data
- **Python** — Research coding
- **PyTorch** — Deep learning
- **Research Writing & Speaking**
- **Leadership & Mentorship**

Presenting at ICML 2026

1. **Kazeev, Nikita**, and Ian Babich. Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments. Forecasting as a New Frontier of Intelligence Workshop at ICML 2026. July 11, 12:15–13:30
2. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. Position: Align AI to Our Aspirations, Not Our Flaws. Pluralistic Alignment Workshop July 11, 15:00–16:30
3. **Kazeev, Nikita** and Ustyuzhanin, Andrey. The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories. Statistical Frameworks for Uncertainty in Agentic Systems Workshop July 11, 15:30–17:00
4. **Kazeev, Nikita** and Ustyuzhanin, Andrey. Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories Structured Probabilistic Inference & Generative Modeling Workshop July 11, 11:40–13:30 & 16:10–17:00

Selected Publications

1. **Kazeev, Nikita**, et al. WyckoffTransformer: Generation of Symmetric Crystals. ICML 2025, May 2025.
2. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). Cherenkov detectors fast simulation using neural networks. *Nuclear Instruments and Methods in Physics Research Section A*. **Alphabetic order, I'm the corresponding author.**